



Protocol for the Examination of Resection Specimens from Patients with Gastrointestinal Stromal Tumor (GIST)

Version: 4.4.0.0

Protocol Posting Date: September 2026

CAP Laboratory Accreditation Program Protocol Required Use Date: June 2027

The changes included in this current protocol version affect accreditation requirements. The new deadline for implementing this protocol version is reflected in the above accreditation date.

For accreditation purposes, this protocol should be used for the following procedures and tumor types:

Procedure	Description
Resection	
Tumor Type	Description
Gastrointestinal stromal tumor	

This protocol is NOT required for accreditation purposes for the following:

Procedure
Biopsy
Local excision
Primary resection specimen with no residual tumor (e.g., following neoadjuvant therapy)
Cytologic specimens
Pediatric GIST, familial GIST (germline mutant <i>KIT</i> or <i>PDGFRA</i>) or syndromic GIST

Version Contributors

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Glossary:

Author: Expert who is a current member of the Cancer Committee, or an expert designated by the chair of the Cancer Committee.

Expert Contributors: Includes members of other CAP committees or external subject matter experts who contribute to the current version of the protocol.

Accreditation Requirements

Synoptic reporting with core and conditional data elements for designated specimen types* is required for accreditation.

- Data elements designated as core must be reported.
- Data elements designated as conditional only need to be reported if applicable.
- Data elements designated as optional are identified with "+". Although not required for accreditation, they may be considered for reporting.

This protocol is not required for recurrent or metastatic tumors resected at a different time than the primary tumor. This protocol is also not required for pathology reviews performed at a second institution (i.e., second opinion and referrals to another institution).

Full accreditation requirements can be found on the CAP website under [Accreditation Checklists](#). A list of core and conditional data elements can be found in the Summary of Required Elements under

Resources on the CAP Cancer Protocols [website](#).

**Includes definitive primary cancer resection and pediatric biopsy tumor types.*

Synoptic Reporting

All core and conditionally required data elements outlined on the surgical case summary from this cancer protocol must be displayed in synoptic report format. Synoptic format is defined as:

- Data element: followed by its answer (response), outline format without the paired Data element: Response format is NOT considered synoptic.
- The data element should be represented in the report as it is listed in the case summary. The response for any data element may be modified from those listed in the case summary, including "Cannot be determined" if appropriate.
- Each diagnostic parameter pair (Data element: Response) is listed on a separate line or in a tabular format to achieve visual separation. The following exceptions are allowed to be listed on one line:
 - Anatomic site or specimen, laterality, and procedure
 - Pathologic Stage Classification (pTNM) elements
 - Negative margins, as long as all negative margins are specifically enumerated where applicable
- The synoptic portion of the report can appear in the diagnosis section of the pathology report, at the end of the report or in a separate section, but all Data element: Responses must be listed together in one location
- Organizations and pathologists may choose to list the required elements in any order, use additional methods in order to enhance or achieve visual separation, or add optional items within the synoptic report. The report may have required elements in a summary format elsewhere in the report IN ADDITION TO but not as replacement for the synoptic report i.e., all required elements must be in the synoptic portion of the report in the format defined above.

Summary of Changes

v 4.4.0.0

- GIST title / lineage is now inclusive of all GIST sites and not limited to stomach
- WHO 6th Edition updates to content and explanatory notes
- Removal of Preresection Treatment question
- Reformatted Tumor Site question
- Updates to Histologic Type question, Mitotic Rate instructional Display Item, Treatment Effect Question, and Molecular Genetic Studies question parenthetical examples
- pTNM Classification section update to include N Suffix conditionally reported question

Reporting Template

Protocol Posting Date: September 2026

Select a single response unless otherwise indicated.

CASE SUMMARY: (GASTROINTESTINAL STROMAL TUMOR (GIST): Resection)

Standard(s): AJCC 8

CLINICAL

+Associated Syndrome

- ☐ Carney triad
- ☐ Carney-Stratakis syndrome
- ☐ Neurofibromatosis type 1
- ☐ Familial GIST syndrome
- ☐ Other (specify): _____
- ☐ Not specified

SPECIMEN

Procedure

- ☐ Local excision
- ☐ Resection (specify type, e.g., partial gastrectomy): _____
- ☐ Other (specify): _____
- ☐ Not specified

TUMOR

Tumor Focality

- ☐ Unifocal
- ☐ Multifocal

Number of Tumors

- ☐ Specify number: _____
- ☐ Other (specify): _____
- ☐ Cannot be determined (explain): _____

Sizes of Tumors: _____

- ☐ Cannot be determined (explain): _____

Multiple Primary Sites (e.g., stomach and small intestine) (required only if applicable)

Please complete a separate checklist for each primary site

- ☐ Not applicable (no additional primary site(s) present)
- ☐ Present: _____

Tumor Site (Note [A](#))

- ☐ Esophagus (specify location): _____
- ☐ Gastroesophageal junction: _____
- ☐ Stomach (specify location): _____
- ☐ Duodenum

- ☐ Jejunum
- ☐ Ileum (excluding ileocecal valve)
- ☐ Meckel diverticulum (site of neoplasm)
- ☐ Small intestine, NOS
- ☐ Appendix: _____
- ☐ Ileocecal valve: _____
- ☐ Cecum
- ☐ Ascending colon
- ☐ Hepatic flexure of colon
- ☐ Transverse colon
- ☐ Splenic flexure of colon
- ☐ Descending colon
- ☐ Sigmoid colon
- ☐ Rectosigmoid junction: _____
- ☐ Rectum: _____
- ☐ Large intestine, NOS
- ☐ Retroperitoneum: _____
- ☐ Peritoneum / abdomen (specify site): _____
- ☐ Other (specify): _____
- ☐ Cannot be determined (explain): _____
- ☐ Not specified

Histologic Type

- ☐ Gastrointestinal stromal tumor, spindle cell type
- ☐ Gastrointestinal stromal tumor, epithelioid type
- ☐ Gastrointestinal stromal tumor, mixed
- ☐ Gastrointestinal stromal tumor, SDH-deficient
- ☐ Gastrointestinal stromal tumor, other (specify): _____

+Histologic Type Comment: _____

Tumor Size

- ☐ Greatest dimension in Centimeters (cm): _____ cm
- ☐ Cannot be determined (explain): _____

Mitotic Rate (Note [B](#))

The mitotic count should be initiated and reported from the area / lesion that on screening magnification reveals the highest level of mitotic activity and be performed as consecutive high-power fields (HPF). The mitotic rate should be determined in 5 mm² of tumor. With the use of older model microscopes, 50 HPF is equivalent to 5 mm². Most modern microscopes with wider fields require approximately 20 to 25 HPF to encompass 5 mm². If necessary, please measure a field of view to accurately determine actual number of fields required to be counted on individual microscopes to encompass 5 mm².

- ☐ Specify mitotic rate per 5 mm²: _____ mitoses per 5 mm²
- ☐ Other (specify): _____
- ☐ Cannot be determined (explain): _____

Histologic Grade (Note [B](#))

- ☐ G1, low-grade (mitotic rate less than or equal to 5 per 5 mm²)
- ☐ G2, high-grade (mitotic rate greater than 5 per 5 mm²)

___ Other (specify): _____
___ GX, cannot be assessed (explain): _____

+Necrosis

___ Not identified
___ Present

+Extent of Necrosis

___ Specify percentage: _____ %
___ Other (specify): _____
___ Cannot be determined: _____
___ Cannot be determined: _____

Treatment Effect (Note [C](#))

___ No known presurgical therapy
___ Not identified
___ Not specified
___ Present

+Percentage of Viable Tumor

___ Specify percentage: _____ %
___ Other (specify): _____
___ Cannot be determined: _____

+Percentage of Necrosis

___ Specify percentage: _____ %
___ Other (specify): _____
___ Cannot be determined: _____
___ Cannot be determined (explain): _____

Risk Assessment (Note [D](#))

___ None
___ Very low risk
___ Low risk
___ Moderate risk
___ High risk
___ Overtly malignant / metastatic
___ Cannot be determined (explain): _____
___ Not applicable

+Tumor Comment: _____

MARGINS

Margin Status

___ All margins negative for GIST

Closest Margin(s) to GIST (select all that apply)

___ Proximal: _____
___ Distal: _____
___ Omental (radial): _____

___ Mucosal: _____
___ Deep: _____
___ Other (specify): _____
___ Cannot be determined (explain): _____

Distance from GIST to Closest Margin

Specify in Centimeters (cm)

___ Exact distance in cm: _____ cm
___ Greater than 1 cm

Specify in Millimeters (mm)

___ Exact distance in mm: _____ mm
___ Greater than 10 mm

Other

___ Other (specify): _____
___ Cannot be determined (explain): _____

___ GIST present at margin

Margin(s) Involved by GIST (select all that apply)

___ Proximal: _____
___ Distal: _____
___ Omental (radial): _____
___ Mucosal: _____
___ Deep: _____
___ Other (specify): _____
___ Cannot be determined (explain): _____
___ Other (specify): _____
___ Cannot be determined (explain): _____
___ Not applicable

+Margin Comment: _____

REGIONAL LYMPH NODES (Note [E](#))

Regional Lymph Node Status

___ Not applicable (no regional lymph nodes submitted or found)
___ Regional lymph nodes present
___ All regional lymph nodes negative for tumor
___ Tumor present in regional lymph node(s)

Number of Lymph Nodes with Tumor

___ Exact number (specify): _____
___ At least (specify): _____
___ Other (specify): _____
___ Cannot be determined (explain): _____
___ Other (specify): _____
___ Cannot be determined (explain): _____

Number of Lymph Nodes Examined

___ Exact number (specify): _____
___ At least (specify): _____

___ Other (specify): _____
___ Cannot be determined (explain): _____

+Regional Lymph Node Comment: _____

DISTANT METASTASIS

Distant Site(s) Involved, if applicable (select all that apply)

___ Not applicable
___ Liver: _____
___ Other (specify): _____
___ Cannot be determined (explain): _____

pTNM CLASSIFICATION (AJCC 8th Edition) (Note [F](#))

Reporting of pT, pN, and (when applicable) pM categories is based on information available to the pathologist at the time the report is issued. As per the AJCC (Chapter 1, 8th Ed.) it is the managing physician's responsibility to establish the final pathologic stage based upon all pertinent information, including but potentially not limited to this pathology report.

Modified Classification (required only if applicable) (select all that apply)

___ Not applicable
___ y (post-neoadjuvant therapy)
___ r (recurrence)

pT Category

___ pT not assigned (cannot be determined based on available pathological information)
___ pT0: No evidence of primary tumor
___ pT1: Tumor 2 cm or less
___ pT2: Tumor more than 2 cm but not more than 5 cm
___ pT3: Tumor more than 5 cm but not more than 10 cm
___ pT4: Tumor more than 10 cm in greatest dimension

T Suffix (required only if applicable)

___ Not applicable
___ (m) Multiple primary synchronous tumors in a single organ

pN Category (Notes [E](#), [F](#))

When no lymph nodes are present (as is often the case with resection for GIST), the pathologic 'N' category is not assigned (pNX is not used for GIST) and should not be reported.

___ pN not assigned (no nodes submitted or found)
___ pN not assigned (cannot be determined based on available pathological information)
___ pN0: No regional lymph node metastasis
___ pN1: Regional lymph node metastasis

N Suffix (required only if applicable) (select all that apply)

___ Not applicable
___ (sn) Sentinel node procedure

___ (f) FNA or core needle biopsy

pM Category (required only if confirmed pathologically) (Notes [E](#), [F](#))

___ Not applicable - pM cannot be determined from the submitted specimen(s)

___ pM1: Distant metastasis

ADDITIONAL FINDINGS

+Additional Findings: _____

SPECIAL STUDIES (Note [G](#))

The CAP GIST Biomarker Template can be used for reporting biomarkers requested.

+Immunohistochemical Studies (select all that apply)

___ Not performed: _____

___ KIT (CD117)

KIT (CD117)

___ Positive

___ Negative

___ Pending

___ DOG1 (ANO1)

DOG1 (ANO1)

___ Positive

___ Negative

___ Pending

___ SDHA

SDHA

___ Intact

___ Deficient

___ Pending

___ SDHB

SDHB

___ Intact

___ Deficient

___ Pending

___ BRAF

BRAF

___ Positive

___ Negative

___ Pending

___ Other (specify): _____

+Molecular Genetic Studies (e.g., KIT, PDGFRA, SDHA / B / C / D, NF1, RAS, or BRAF mutational analysis, or NTRK, BRAF, FGFR1 or other fusion gene analysis)

___ Performed, see biomarker report: _____

___ Performed (specify method(s) and result(s)): _____

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___ Pending
___ Not performed

COMMENTS

Comment(s): _____

Explanatory Notes

A. Location

Gastrointestinal stromal tumors may occur anywhere along the entire length of the tubal gut, as well as in extraviseral locations, which include the omentum, mesentery, pelvis, and retroperitoneum.^{1,2,3,4,5,6} Typically, these tumors arise from the wall of the gut and extend inward toward the mucosa, outward toward the serosa, or in both directions. Lesions that involve the wall of the gastrointestinal (GI) tract frequently cause ulceration of the overlying mucosa. Infrequently, lesions invade through the muscularis mucosa to involve the mucosa. Mucosal invasion is an adverse prognostic factor in numerous studies. Because the anatomic location along the GI tract affects prognosis, with location in the stomach having a more favorable prognosis, it is very important to specify anatomic location as precisely as possible.^{5,6}

References

1. Fletcher CD, Berman JJ, Corless C, et al. Diagnosis of gastrointestinal stromal tumors: a consensus approach. *Hum Pathol*. 2002;33(5):459-465.
2. Miettinen M, Lasota J. Gastrointestinal stromal tumors: definition, clinical, histological, immunohistochemical, and molecular genetic features and differential diagnosis. *Virchows Arch*. 2001;438(1):1-12.
3. Reith JD, Goldblum JR, Lyles RH, Weiss SW. Extragastric (soft tissue) stromal tumors: an analysis of 48 cases with emphasis on histologic predictors of outcome. *Mod Pathol*. 2000;13(5):577-585.
4. WHO Classification of Tumours Editorial Board. *Digestive system tumours*. Lyon (France): International Agency for Research on Cancer; 2026. (WHO classification of tumours series, 6th ed.; vol. 1).
5. Miettinen M, Sobin LH, Lasota J. Gastrointestinal stromal tumors of the stomach: a clinicopathologic, immunohistochemical, and molecular genetic study of 1765 cases with long-term follow-up. *Am J Surg Pathol*. 2005;29(1):52-68.
6. Miettinen M, Felisiak-Golabek A, Wang Z, Inaguma S, Lasota J. GIST Manifesting as a Retroperitoneal Tumor: Clinicopathologic Immunohistochemical, and Molecular Genetic Study of 112 Cases. *Am J Surg Pathol*. 2017 May;41(5):577-585. PMID: 28288036

B. Histologic Grade

Histologic grading in GIST, unlike in soft tissue sarcoma, only takes mitotic rate into account. GIST is generally less proliferative than many other soft tissue tumors and the threshold for separating low from high-grade tumors occurs at 5 mitotic figures per 5 mm².^{1,2,3}

GX: Grade cannot be assessed

G1: Low grade; mitotic rate $\leq 5/5$ mm²

G2: High grade; mitotic rate $>5/5$ mm²

The mitotic count should be initiated in an area that on screening magnification reveals the highest level of mitotic activity and be performed as consecutive high-power fields (HPF). Stringent criteria should be applied when counting mitotic figures; pyknotic or apoptotic nuclei should not be regarded as mitosis.

Note: Mitoses should be counted in 5 mm² of tumor.^{2,3} With the use of older model microscopes, 50 HPF is equivalent to 5 mm². Most modern microscopes with wider fields require approximately 20 to 25 HPF to

encompass 5 mm². If necessary, please measure a field of view to accurately determine an actual number of fields required to be counted on individual microscopes to encompass 5 mm².

References

1. Amin MB, Edge SB, Greene FL, et al, eds. *AJCC Cancer Staging Manual*. 8th ed. New York, NY: Springer; 2017.
2. Miettinen M, Lasota J. Gastrointestinal stromal tumors: pathology and prognosis at different sites. *Semin Diagn Pathol*. 2006;23(2):70-83.
3. WHO Classification of Tumours Editorial Board. *Digestive system tumours*. Lyon (France): International Agency for Research on Cancer; 2026. (WHO classification of tumours series, 6th ed.; vol. 1).

C. Treatment Effect

Gastrointestinal stromal tumors respond well to the newer targeted systemic therapies, imatinib mesylate, and sunitinib malate. The types of treatment effects that have been seen are hypocellularity, myxoid stroma, fibrosis, and necrosis. Nests of viable tumor cells are virtually always seen. Because all of these histologic features can be seen in untreated GIST, it is not possible to know whether they are due to treatment or not. As a practical compromise, it is best to report the percentage of viable tumor after treatment.

D. Risk Assessment

Biopsies are suboptimally positioned for GIST risk stratification as these may not include sufficient tumor (i.e., 5 mm²) for mitotic counting and may not sample mitotic “hot spots”. Furthermore, the risk for metastasis or tumor related death presumes that the GIST has been removed. On biopsy, one may attempt to risk stratify a GIST, using location, available material for mitotic count, and clinicoradiologic size into account.

Biopsies are more predictive if overtly high mitotic count/high grade/high risk on biopsy, based on mitoses and clinicoradiologic size yet low mitotic count/low grade on biopsy may underestimate actual mitoses on resection and not be accurate due to sampling. Because GIST can recur many years after initial excision, most GIST is now regarded as having at least some potential for distant metastasis. This concept was originally the result of a National Cancer Institute-sponsored consensus conference that was held in 2002.¹ More specific data generated by large follow-up studies refined the biologic potential assessment.^{2,3,4,5,6,7} Criteria obtained from those data were adopted in a National Cancer Care Network (NCCN) Task Force report on GIST.⁸ We have adopted the criteria for risk stratification, as indicated in Table 1.^{2,3,4,5,6,7} The scheme includes anatomic site as a factor because small bowel GIST carries a higher risk of progression than gastric GIST of similar size and mitotic activity. This prognostic assessment applies best to *KIT/PDGFR*A mutant GIST whereas SDH-deficient GIST are more unpredictable.⁹ For anatomic sites not listed in this table, such as esophagus, mesentery, and peritoneum, or in the case of “insufficient data,” it is best to use risk criteria for jejunum/ileum.¹⁰

Table 1. Guidelines for Risk Assessment of Primary Gastrointestinal Stromal Tumor (GIST)

Tumor Parameters		Risk of Progressive Disease#(%)			
Mitotic Rate	Size	Gastric	Duodenum	Jejunum/Ileum	Rectum
≤5 per 5 mm ²	≤2 cm	None (0%)	None (0%)	None (0%)	None (0%)

	>2 - ≤5 cm	Very low (1.9%)	Low (8.3%)	Low (4.3%)	Low (8.5%)
	>5 - ≤10 cm	Low (3.6%)	(Insufficient data##)	Moderate (24%)	(Insufficient data##)
	>10 cm	Moderate (12%)	High (34%)	High (52%)	High (57%)
>5 per 5 mm ²	≤2 cm	None	(Insufficient data##)	High	High (54%)
	>2 - ≤5 cm	Moderate (16%)	High (50%)	High (73%)	High (52%)
	>5 - ≤10 cm	High (55%)	(Insufficient data##)	High (85%)	(Insufficient data##)
	>10 cm	High (86%)	High (86%)	High (90%)	High (71%)

Adapted with permission from Miettinen and Lasota.⁵ Copyright 2006 by Elsevier.

Data based on long-term follow-up of 1055 gastric, 629 small intestinal, 144 duodenal, and 111 rectal GIST from the pre-imatinib era.^{2,3,4,6}

Defined as metastasis or tumor-related death

Denotes small number of cases

Note: See Note B, "Histologic Grade," regarding the number of high-power fields to evaluate.

References

1. Fletcher CD, Berman JJ, Corless C, et al. Diagnosis of gastrointestinal stromal tumors: a consensus approach. *Hum Pathol*. 2002;33(5):459-465.
2. Miettinen M, Sobin LH, Lasota J. Gastrointestinal stromal tumors of the stomach: a clinicopathologic, immunohistochemical, and molecular genetic study of 1765 cases with long-term follow-up. *Am J Surg Pathol*. 2005;29(1):52-68.
3. Miettinen M, Furlong M, Sarlomo-Rikala M, Burke A, Sobin LH, Lasota J. Gastrointestinal stromal tumors, intramural leiomyomas, and leiomyosarcomas in the rectum and anus: a clinicopathologic, immunohistochemical, and molecular genetic study of 144 cases. *Am J Surg Pathol*. 2001;25(9):1121-1133.
4. Miettinen M, Kopczynski J, Makhlof HR, et al. Gastrointestinal stromal tumors, intramural leiomyomas, and leiomyosarcomas in the duodenum: a clinicopathologic, immunohistochemical, and molecular genetic study of 167 cases. *Am J Surg Pathol*. 2003;27(5):625-641.
5. Miettinen M, Lasota J. Gastrointestinal stromal tumors: pathology and prognosis at different sites. *Semin Diagn Pathol*. 2006;23(2):70-83.
6. Miettinen M, Makhlof H, Sobin LH, Lasota J. Gastrointestinal stromal tumors of the jejunum and ileum: a clinicopathologic, immunohistochemical, and molecular genetic study of 906 cases before imatinib with long-term follow-up. *Am J Surg Pathol*. 2006;30(4):477-489.
7. Miettinen M, Felisiak-Golabek A, Wang Z, Inaguma S, Lasota J. GIST Manifesting as a Retroperitoneal Tumor: Clinicopathologic Immunohistochemical, and Molecular Genetic Study of 112 Cases. *Am J Surg Pathol*. 2017 May;41(5):577-585. PMID: 28288036.
8. Demetri GD, Benjamin RS, Blanke CD, et al; NCCN Task Force. NCCN Task Force report: management of patients with gastrointestinal stromal tumor (GIST)--update of the NCCN clinical practice guidelines. *J Natl Compr Canc Netw*. 2007;5(Suppl 2): S1-S29.
9. WHO Classification of Tumours Editorial Board. *Digestive system tumours*. Lyon (France): International Agency for Research on Cancer; 2026. (WHO classification of tumours series, 6th ed.; vol. 1).

10. Hu, S., Alpert, L., Cates, J. M. M., et al. Gastrointestinal stromal tumors (GISTs) arising in uncommon locations: clinicopathologic features and risk assessment of esophageal, colonic, and appendiceal GISTs. *Mod Pathol.* 2022; 35(4), 554 -563.

E. Regional Lymph Nodes, Metastasis

Gastrointestinal stromal tumors generally metastasize to a very limited subset of anatomic sites.¹ These tumors rarely metastasize to lymph nodes, which is important to note because lymphadenectomy is unnecessary except in rare circumstances when an enlarged or otherwise suspicious lymph node is encountered. Gastrointestinal stromal tumors metastasize predominantly to the liver or to the peritoneal surfaces, where there can be disseminated intra-abdominal disease presenting as innumerable metastatic nodules. Very rarely, GIST metastasize to the lungs. This situation is associated with rectal location or very advanced disease.² Metastasis to bone has also been documented, but it is very rare.

References

1. Fletcher CD, Berman JJ, Corless C, et al. Diagnosis of gastrointestinal stromal tumors: a consensus approach. *Hum Pathol.* 2002;33(5):459-465.
2. Miettinen M, Furlong M, Sarlomo-Rikala M, Burke A, Sobin LH, Lasota J. Gastrointestinal stromal tumors, intramural leiomyomas, and leiomyosarcomas in the rectum and anus: a clinicopathologic, immunohistochemical, and molecular genetic study of 144 cases. *Am J Surg Pathol.* 2001;25(9):1121-1133.

F. pTNM CLASSIFICATION

The American Joint Committee on Cancer (AJCC) and International Union Against Cancer (UICC) GIST staging system is recommended.¹ The staging system should not be applied to pediatric GIST, familial GIST (germline mutant *KIT* or *PDGFRA*) or syndromic GIST (GIST arising in the setting of neurofibromatosis type 1, Carney triad, or Carney dyad also known as Carney-Stratakis syndrome).

TNM Descriptors

For identification of special cases of TNM or pTNM classifications, the “m” suffix and “y” and “r” prefixes are used. Although they do not affect the stage grouping, they indicate cases needing separate analysis.

The “m” suffix indicates the presence of multiple primary tumors in a single site and is recorded in parentheses: pT(m)NM.

The “y” prefix indicates those cases in which classification is performed during or after initial multimodality therapy (i.e., neoadjuvant chemotherapy, radiation therapy, or both chemotherapy and radiation therapy). The cTNM or pTNM category is identified by a “y” prefix. The ycTNM or ypTNM categorizes the extent of tumor actually present at the time of that examination. The “y” categorization is not an estimate of tumor before multimodality therapy (i.e., before initiation of neoadjuvant therapy).

The “r” prefix indicates a recurrent tumor when staged after a documented disease-free interval and is identified by the “r” prefix: rTNM.

T Category Considerations

In the case of ruptured tumors, estimates of tumor size can be obtained from radiologic data, if available.

N Category Considerations

Regional nodal metastasis is extremely rare in GIST, and there is no routine indication for lymph node biopsy or lymph node dissection. When no lymph nodes are resected or present in the specimen (as is often the case with resections for GIST), the pathologic 'N' category is not assigned; pNX should not be used.

M Category Considerations

Most GIST metastasize to intra-abdominal soft tissues, liver, or both. Intra-abdominal metastasis refers to tumor involvement in the abdominal cavity away from the primary mass. Such metastasis is usually to the serosal surfaces of the abdomen, pelvis, and retroperitoneum. Multiple primary tumors can be seen in the setting of neurofibromatosis type 1 or familial GIST syndrome and should not be considered intra-abdominal metastasis. Rare cases of multiple independent GIST at different GI locations have been reported. In the absence of a primary gastrointestinal GIST, solitary omental, mesenteric, pelvic, or retroperitoneal GIST should be considered primary tumors because extra-gastrointestinal GIST has been described. Liver metastasis implies the presence of metastatic tumor inside the liver parenchyma as one or more nodules. Adherence to liver capsule, even if extensive, as sometimes seen in gastric GIST, should not be considered liver metastasis.

Stage Groupings:

Although T, N, and M definitions are identical for all GIST, separate stage grouping schemes are provided for gastric and small intestinal tumors. Primary omental GIST should follow the gastric GIST staging group scheme. GIST arising in other locations (i.e., mesentery, esophagus, colon, and rectum) are to follow the small intestinal group staging scheme upstaging scheme.

References

1. Amin MB, Edge SB, Greene FL, et al, eds. *AJCC Cancer Staging Manual*. 8th ed. New York, NY: Springer; 2017.

G. Ancillary Studies

Immunohistochemical Analysis

Because small-molecule kinase inhibitor therapy is highly effective in the treatment of GIST, it has become imperative to distinguish GIST from histologic mimics, mainly leiomyoma, leiomyosarcoma, schwannoma, and desmoid fibromatosis.^{1,2} Immunohistochemistry is instrumental in the workup of GIST. For the initial workup of GIST, a basic immunohistochemical panel including CD117 (KIT), DOG1 (ANO1), desmin, and S100 protein is recommended. GIST is immunoreactive for KIT (CD117) (95%) and/or DOG1 (>99%).^{3,4,5} KIT immunoreactivity is usually strong and diffuse but can be more focal in unusual cases (Figure 1, A, and B). It is not unusual for GIST to exhibit dot-like perinuclear staining (Figure 1, C), while less commonly, cases exhibit membranous staining (Figure 1, D). These patterns do not correlate with mutation type or response to therapy. Most KIT-negative/DOG1 positive GIST are gastric or extra-visceral and almost invariably harbor a *PDGFRA* mutation.⁶ DOG1 expression is not related to mutational status, and is a useful marker to identify KIT-negative GISTs.^{4,5} Approximately 70% of GISTs are positive for CD34, 30% to 40% for smooth muscle actin, 5% for S100 protein (usually focal), 5% for desmin (usually focal), and 1% to 2% for keratin (weak/focal).⁷ PanTrk immunohistochemistry may be positive in GIST independent of *NTRK* fusion, and this immunostain is not recommended in GIST.

Succinate dehydrogenase (SDH) is a mitochondrial enzyme comprising four subunits (SDHA, SDHB, SDHC, and SDHD) that is involved in the Krebs cycle. Genetic alteration of any of the four subunits results in SDH dysfunction and subsequent loss of SDHB expression by immunohistochemistry. SDH-deficient GISTs are characterized by early onset, gastric location, female predominance, multifocality, a plexiform/multinodular growth pattern, frequent lymph node metastases, and epithelioid morphology; they are *KIT* and *PDGFRA* wild-type.^{8,9} One-half to two-thirds of SDH-deficient GISTs are associated with a pathogenic germline mutation in an *SDH* subunit gene (*SDHx*), most commonly *SDHA*. A substantial proportion of the remaining tumors demonstrate *SDHC* promoter hypermethylation, particularly in the setting of Carney triad. Rare tumors harbor biallelic somatic *SDHx* alterations or remain molecularly unexplained despite comprehensive testing. Thus, it is recommended that all gastric GISTs be screened for loss of SDH by immunohistochemistry, best accomplished by immunostaining for SDHB, which is lost in all SDH-deficient GIST, independent of the SDH-subunit that is inactivated.^{10,11,12,13} Mutations in *SDHA* are detected in ~30% of SDH-deficient GISTs and loss of expression of SDHA specifically identifies tumors with *SDHA* mutations; other SDH-deficient GISTs show normal (intact) cytoplasmic staining.^{14,15} Patients with SDH-deficient GIST should be referred to a genetic counselor for appropriate workup.

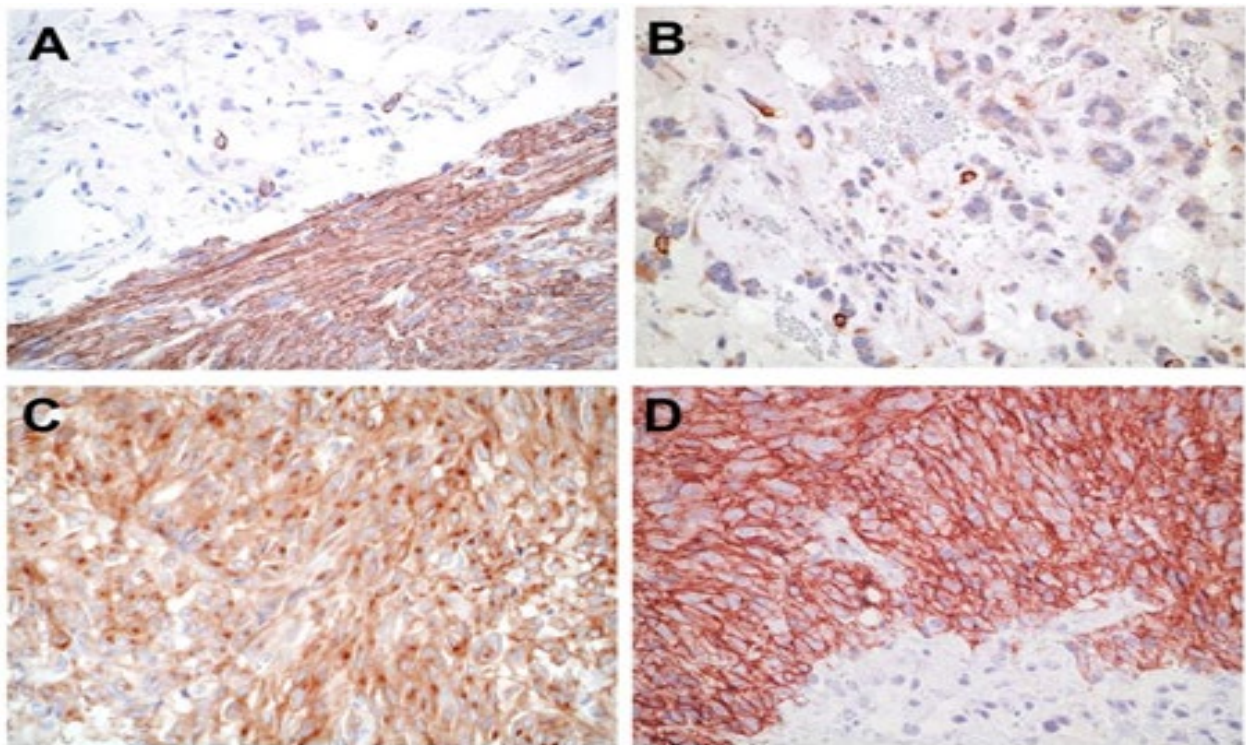


Figure 1. Patterns of KIT staining in gastrointestinal stromal tumor (GIST). A. Diffuse and strong immunoreactivity in a typical GIST. B. Focal and weak staining in an epithelioid gastric GIST with a *PDGFRA* mutation. C. Dot-like perinuclear staining. D. Membranous pattern staining. (Original magnification X400)

Molecular Analysis

Approximately 75% of GISTs possess activating mutations in *KIT*, whereas another 10% have activating mutations in *PDGFRA*.^{16,17,18,19} The resulting mutant proteins demonstrate ligand-independent activation. *KIT* and *PDGFRA* each contain 21 exons. However, mutations cluster within “hotspots”: exons 9, 11, 13, and 17 in *KIT* and exons 12, 14, and 18 in *PDGFRA* (Figure 2). About 10 to 15% of GISTs appear to be negative for both *KIT* and *PDGFRA* mutations. The most recent NCCN Task Force on GIST strongly encouraged that *KIT* and *PDGFRA* mutational analysis be performed in unresectable and metastatic GISTs for which use of tyrosine kinase inhibitors (TKIs) is considered; mutation testing should also be considered for patients with primary disease, particularly if high-risk. *KIT* and *PDGFRA* mutation status can be readily determined from paraffin-embedded tissue. Secondary (acquired) resistance mutations can develop in the setting of long-term imatinib mesylate treatment. These are usually point mutations that occur most commonly in *KIT* exons 13, 14, 17, and 18.²⁰

Recent studies focusing on the molecular classification of GIST recognized two major subgroups: succinate dehydrogenase (SDH)-competent (92-95%) and SDH-deficient GIST (5-8%), both of which can arise in the sporadic or familial setting.^{10,11} SDH-competent GISTs include tumors with mutations of *KIT* or *PDGFRA*, as well as those with mutations in *NF1* (2-6%) or *BRAF* ($\leq 1\%$) and rarely fusions involving *FGFR1* or *BRAF*.^{21,22,23,24,25,26} Other rare molecular genetic events have been described including *PIK3CA* mutation (which is not necessarily independent of *KIT*/*PDGFRA* activation), *NTRK* fusion (some of these tumors may be better characterized as non-GISTs), and *RAS* family genes (very uncommon).

SDH-deficient GISTs are characterized by genetic or epigenetic inactivation of any SDH subunit. As mentioned previously, up to two-thirds of SDH-deficient GISTs are associated with a pathogenic germline mutation in an SDH subunit gene (*SDHx*), most commonly *SDHA*, with a substantial proportion of the remaining tumors demonstrating *SDHC* promoter hypermethylation. Somatic biallelic inactivation of an *SDHx* gene is considered rare.^{27,28,29} The majority of pediatric GIST arise in Carney triad (SDH-deficient GIST, pulmonary chondroma, extra-adrenal paraganglioma; typically due to *SDHC* promoter methylation) or Carney-Stratakis syndrome (SDH-deficient GIST and paraganglioma due to germline mutation in an *SDHx* gene). Germline mutations in *SDHx* can also lead to SDH-deficient renal cell carcinomas and pituitary tumors.

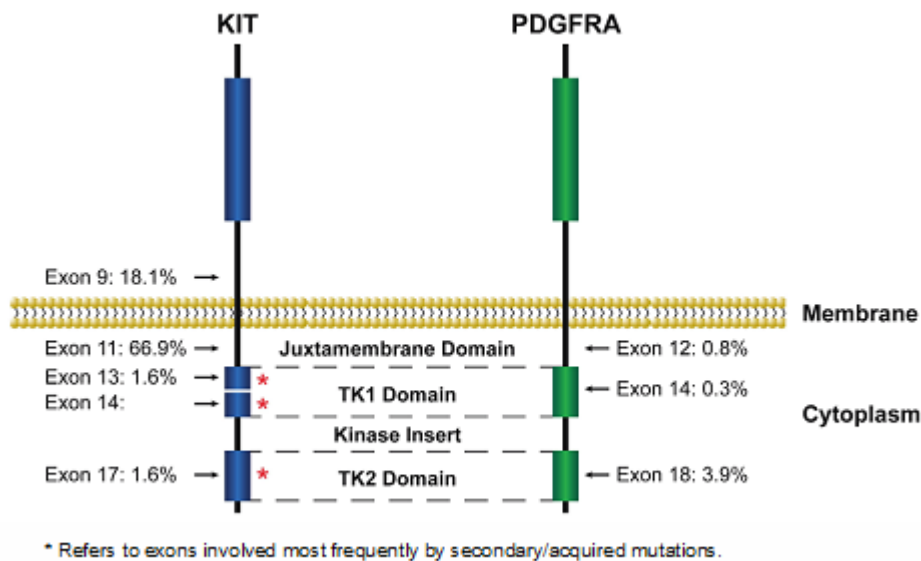


Figure 2. Locations and frequency of activating *KIT* and *PDGFRA* mutations in GIST. Adapted with permission from Heinrich et al.¹⁶ Copyright 2003 by the American Society of Clinical Oncology. All rights reserved.

KIT and *PDGFRA* are excellent targets for small-molecule tyrosine kinase inhibitors (TKIs), and compounds of this class, including imatinib mesylate, sunitinib malate, avapritinib (specifically indicated in *PDGFRA* D842V-mutant tumors), regorafenib, and ripretinib, are Food and Drug Administration-approved for the treatment of GIST.^{30,31,32,33} SDH-deficient GISTs are usually resistant to imatinib but may respond to sunitinib.^{10,34} Clinical interest in mutation status is driven by differential response to TKIs, which may drive drug selection.³⁵ Detection of secondary (acquired) resistance mutations may also impact drug selection as their significance is further defined.

NTRK fusions in GIST are very uncommon. 10 to 15% of tumors may stain positively for pan-TRK, reflecting either wild-type TRK protein expression or non-specific staining, and NGS or FISH are preferred to immunohistochemistry for the detection of *NTRK* fusions in GIST.^{36,37,38}

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